

## Hyperparameter Tuning Experiments- Ineza Bella Melissa

| Hyperparameter Set                                                                                  | Noted Behavior                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| lr=1e-4, gamma=0.99, batch=32, epsilon_start=1.0, epsilon_end=0.01, epsilon_decay=0.1 (baseline)    | Baseline configuration. Log data for this run was incomplete in TensorBoard (only a partial ~99k-step record was found) — recommend re-verifying this run's full 300k-step curve before final submission.                                                                                |
| lr=1e-3, gamma=0.99, batch=32, epsilon_start=1.0, epsilon_end=0.01, epsilon_decay=0.1               | Worst performer (tied), final reward -21.00. Learning rate 10x higher than baseline caused overly aggressive updates — consistent with the classic failure mode of an unstable/too-high learning rate preventing convergence.                                                            |
| lr=2.5e-5, gamma=0.99, batch=32, epsilon_start=1.0, epsilon_end=0.01, epsilon_decay=0.1             | Final reward -20.99. Learning rate lower than baseline led to very slow learning — not enough training steps (300k) for improvement to show yet.                                                                                                                                         |
| lr=1e-4, gamma=0.90, batch=32, epsilon_start=1.0, epsilon_end=0.01, epsilon_decay=0.1               | 2nd best performer, final reward -20.50. Lower gamma (more myopic agent) performed better than baseline — likely because Pong rewards arrive quickly, so discounting the distant future less aggressively reduced noisy Q-value estimates early in training.                             |
| lr=1e-4, gamma=0.995, batch=32, epsilon_start=1.0, epsilon_end=0.01, epsilon_decay=0.1              | Final reward -20.85, close to baseline. A longer planning horizon (higher gamma) neither helped nor hurt noticeably at this timestep count.                                                                                                                                              |
| lr=1e-4, gamma=0.99, batch=16, epsilon_start=1.0, epsilon_end=0.01, epsilon_decay=0.1               | Final reward -20.94, worse than baseline. Smaller batch size produced noisier gradient updates than the default, slowing stable learning.                                                                                                                                                |
| lr=1e-4, gamma=0.99, batch=128, epsilon_start=1.0, epsilon_end=0.01, epsilon_decay=0.1              | BEST performer overall, final reward -19.85. Quadrupling the batch size gave the most stable gradient updates of all 10 configs, which translated into the fastest learning progress. This config was selected for extended training (1,000,000 timesteps) and the final gameplay video. |
| lr=1e-4, gamma=0.99, batch=32, epsilon_start=1.0, epsilon_end=0.1, epsilon_decay=0.1                | Final reward -20.91, slightly worse than baseline. Higher epsilon_end means the agent kept acting randomly more often even late in training, slightly reducing its ability to exploit what it had learned.                                                                               |
| lr=1e-4, gamma=0.99, batch=32, epsilon_start=1.0, epsilon_end=0.01, epsilon_decay=0.3 (slow decay)  | Worst performer (tied), final reward -21.00. Exploration rate stayed high for 30% of training instead of the default 10%, so by 300k steps the agent had spent much less time exploiting its learned policy — performance looked close to random.                                        |
| lr=1e-4, gamma=0.99, batch=32, epsilon_start=1.0, epsilon_end=0.01, epsilon_decay=0.02 (fast decay) | Final reward -20.98, worse than baseline. Epsilon dropped to its minimum almost immediately (2% of training), so the agent locked into a mostly-random early policy too soon, before it had explored enough to learn a good one.                                                         |